

Erdős Problem #687

EP-687

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $Y(x)$ be the maximal y such that there exists a choice of congruence classes a_p for all primes $p \leq x$ such that every integer in $[1, y]$ is congruent to at least one of the $a_p \pmod{p}$.

Give good estimates for $Y(x)$. In particular, can one prove that $Y(x) = o(x^2)$ or even $Y(x) \ll x^{1+o(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>